

CALCULATION OF FAST NEUTRON REMOVAL CROSS-SECTIONS FOR DIFFERENT SHIELDING MATERIALS

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ABSTRACT

In order to help designer of nuclear technology, it is useful to calculate the macroscopic effective removal cross-sections (Σ_R) for fast neutrons. This parameter is used to characterize the attenuation of fast neutrons in materials. In this paper, the macroscopic effective removal cross-sections (Σ_R) for fast neutrons have been calculated theoretically for the following shielding materials: Pure polyethylene, 1 % Borated Polyethylene, 5% Borated Polyethylene, 5.45 % Borated Polyethylene, 8.97% Borated Polyethylene, 30% Borated Polyethylene, 7.5% Lithium Polyethylene, 78.5% Bismuth-loaded Polyethylene, 90% Bismuth-loaded Polyethylene, Borated Silicone, Flexi-Boron Shielding, Borated Hydrogen-loaded castable dry mix, Borated hydrogenated mix, Borated-lead Polyethylene, K-resin, resin 250WD, SUS304, Krafton-HB and Premadex. The results obtained can be used to select the most effective shielding material.

KEYWORDS: Shielding Materials, Neutron, Removal Cross-Section

INTRODUCTION

Nuclear technology is used in several fields such as industry, medicine, agriculture and scientific research and has many advantages but it is dangerous. In fact, this technology is based on ionizing radiation which has harmful effects on human health and environment. Therefore, it was necessary to evaluate the risks and quantify the level of exposure to such radiations and develop technologies for protecting against these radiations.

Radiation shielding involves at placing a shielding material between the ionizing radiations source and the worker or the environment. The radiations which have to be considered are: x and gamma rays, alpha particles, beta particles, and neutrons, each type of these radiations interacts in different ways with shielding material. Therefore, the effectiveness of shielding varies with the type and energy of radiation and also varies with the used shielding material.

The best materials for protection against ionizing radiation are mixture of hydrogenous materials (polyethylene, water and many plastics) and neutron absorbing elements (B, Li, Bi, Cl, etc.), because they reduce both the intensity of gamma rays and neutrons, indeed, hydrogen slows fast and intermediate neutrons energy via inelastic scattering, and they become thermal neutrons which are absorbed by neutron absorbing elements which have a very high neutron absorption cross-section.

Neutron penetration in shielding is characterized by several parameters such as the effective removal cross-section, the macroscopic thermal neutron cross section. In this study, the macroscopic effective removal cross-section of fast neutrons is calculated theoretically.

Materials used in this study are used in neutron shielding applications.

Pure polyethylene, 1 % Borated Polyethylene, 5% Borated Polyethylene, 5.45 % Borated Polyethylene, 8.97% Borated Polyethylene, 30% Borated Polyethylene, 7.5% Lithium Polyethylene, 78.5% Bismuth-loaded Polyethylene, 90% Bismuth-loaded Polyethylene, Borated Silicone, Flexi-Boron Shielding, Borated Hydrogen-loaded castable dry mix and Borated hydrogenated mix are manufactured by a commercial company (Bladewerx) and are available under the Shieldwerx™ trade name (Bladewerx company web page: <http://www.shieldwerx.com>). And Borated-lead Polyethylene, K-resin, resin 250WD, SUS304, Krafton-HB and Premadex are used in the following references respectively (S. C. Gujrathi and J. M. D'Auria (1972); H.Y. Kang et al. 2008; A.M. Sukegawa et al. 2011; J. G. Fantidis et al. 2010). The chemical compositions of these materials are listed in Tables 1-19.

METHODOLOGY

Neutrons are electrically neutral particles, during their passage through a material medium, they interact with the nuclei of atoms in two ways, either by diffusion or absorption. The interaction of neutrons with the atoms described by the total microscopic cross-section σ_t , expresses the probability that a neutron of a given energy interacts with the atoms of the traversed material and it is defined as the sum of the microscopic cross section scattering σ_s and the microscopic cross-section absorption σ_a .

$$\sigma_t = \sigma_s + \sigma_a \quad (1)$$

The attenuation of neutrons during their passage through material medium depends not only on the microscopic cross-section but also on the number of nuclei within this environment. The physical quantity bound these two parameters, called total macroscopic cross-section denoted Σ_t and defined by (J.E. Martin 2000; J.K. Shultis et al. 2008; J.K. Shultis et al. 1996):

$$\Sigma_t = \frac{\rho N_A}{A} \sigma_t \quad (2)$$

Where ρ is the density (g cm^{-3}), N_A is Avogadro's Number and A is the atomic mass.

Σ has the dimensions of the inverse of the length, their unit is cm^{-1} .

In the same way as a beam of photons, when the parallel beam of monoenergetic neutrons passes through a material medium, it will be attenuated due to absorption and scattering. The attenuation of neutrons in matter follows the following law (J.E. Martin 2000; J.K. Shultis et al. 2008; J.K. Shultis et al. 1996):

$$I = I_0 e^{-\Sigma_t x} \quad (3)$$

Where I_0 and I are respectively the intensities of neutrons unmitigated and mitigated, x (cm) is the thickness of the material medium and Σ_t represents the total macroscopic cross-section.

So the case of fast neutron attenuation is described by another parameter called the "removal cross-section", denoted by Σ_R (cm^{-1}) and is different from the total macroscopic cross-section but it has a fraction of it. The removal cross-section presents the probability that a fast or fission-energy neutron undergoes a first collision, which removes it from the group of penetrating uncollided neutrons (E.P. Blizard et al. 1962; J.J. Duderstadt et al. 1976). Indeed, in the MeV-energy region, the absorption cross-section of neutrons is very low compared to the scattering cross-section. In fact, the fast neutrons are not directly absorbed during their passage through the shielding hydrogenated, but they slow primarily by

successive elastic collisions with the nuclei of light elements and when their energy is in the order of the thermal energy (0.025 eV), they are absorbed by the nuclei of heavy elements via interaction radiative capture (A.B. Chilton et al. 1984).

For energies between 2 and 12 MeV, the effective removal cross-section will be almost constant and when the traversed medium contains a large amount of hydrogen $\Sigma_R = \Sigma_t$ and when materials contain a small fraction of hydrogen $\Sigma_R = \frac{2}{3}\Sigma_t$ for energy between 6-8 MeV (M.F. Kaplan 1989; S. Glasstone et al. 1986; A.E. Profio 1979; J. Wood 1982).

Generally, shielding materials are chemical compounds or mixtures, their macroscopic removal cross-section is calculated from the value of Σ_R of their constituent elements and it is given by the following formula (M.F. Kaplan 1989; S. Glasstone et al. 1986; A.E. Profio 1979; J. Wood 1982):

$$\Sigma_R = \sum_i W_i (\Sigma_R / \rho)_i \quad (4)$$

Where W_i , ρ and $(\Sigma_R / \rho)_i$ are respectively the partial density (g cm^{-3}), density and mass removal cross section of the i th constituent. The values of Σ_R / ρ ($\text{cm}^2 \text{g}^{-1}$) of all the elements which constitute the shielding materials used in this study were taken from (A.B. Chilton et al. 1984; M.F. Kaplan 1989; A.E. Profio 1979; J. Wood 1982; A.M. El-Khayatt 2010; A.M. El-Khayatt et al. 2009).

In this study, the effective removal cross-section (Σ_R) of fast neutrons has been calculated for different shielding materials by using formula (4). The elemental composition of materials used in this work, its fractions by weight, partial densities, values of W_i , $(\Sigma_R / \rho)_i$ and calculated (Σ_R) values are listed in Table 1-19.

RESULTS AND DISCUSSIONS

The elemental composition, its fraction by weight, the mass removal cross-section (Σ_R / ρ), the partial density ρ and the macroscopic effective removal cross-section (Σ_R) of fast neutrons for Pure polyethylene, 1 % Borated Polyethylene, 5% Borated Polyethylene, 5.45 % Borated Polyethylene, 8.97% Borated Polyethylene, 30% Borated Polyethylene, 7.5% Lithium Polyethylene, 78.5% Bismuth-loaded Polyethylene, 90% Bismuth-loaded Polyethylene, Borated Silicone, Flexi-Boron Shielding, Borated Hydrogen-loaded castable dry mix, Borated hydrogenated mix, Borated-lead Polyethylene, K-resin, resin 250WD, SUS304, Krafon-HB and Premadex, are listed respectively in Tables 1-19. It can be seen from these tables that the section depends on the elemental composition and density of materials.

Also, it can be noted that the contribution of the light elements to the total removal cross-section is more important compared to the heavy elements. This is due to the fact that the light elements have very high mass removal cross-section (Σ_R / ρ) and especially hydrogen, therefore when its mass fraction increases, their contribution to the total removal cross-section increases and vice versa.

However, the maximum value of (Σ_R) has been observed for the K-resin whereas 90% Bismuth-loaded Polyethylene has the minimum value.

The higher concentration of hydrogen (14.86% H and 30%B) in the chemical composition of the K-resin in comparison to that of the other materials and their high density explain why the K-resin has the maximum value. And the 90% Bismuth-loaded Polyethylene has the minimum value because it contains high concentration of bismuth (90%) which has a very small mass removal cross-section (Σ_R / ρ) compared to other elements.

Table 1: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Pure Polyethylene ($\rho = 0.92 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.5980	0.1437	0.1322	0.0791
C	0.0502	0.8563	0.7878	0.0396
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.1186

Table 2: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 1 % Borated Polyethylene ($\rho = 1.7 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0584	0.0993	0.0594
B	0.0575	0.01	0.0170	0.0010
C	0.0502	0.1802	0.3063	0.0154
O	0.0405	0.4783	0.8131	0.0329
Na	0.0341	0.0019	0.0032	0.0001
Mg	0.0333	0.0014	0.0024	0.0001
Al	0.0293	0.2494	0.4240	0.0124
Si	0.0252	0.0026	0.0044	0.0001
S	0.0277	0.0002	0.0003	0.0000
Ca	0.0243	0.0153	0.0260	0.0006
Fe	0.0214	0.0002	0.0003	0.0000
Sr	0.016	0.001	0.0017	0.0000
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.1221

Table 3: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 5% Borated Polyethylene ($\rho = 0.95 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.116	0.1102	0.0659
B	0.0575	0.05	0.0475	0.0028
C	0.0502	0.612	0.5814	0.0292
O	0.0405	0.222	0.2109	0.0086
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.1064

Table 4: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 5.45 % Borated Polyethylene ($\rho = 1.6 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0572	0.0915	0.0547
B	0.0575	0.0545	0.0872	0.0050
C	0.0502	0.2596	0.4154	0.0209
O	0.0405	0.3969	0.6350	0.0257
Na	0.0341	0.0023	0.0037	0.0001
Mg	0.0333	0.0076	0.0122	0.0004
Al	0.0293	0.1192	0.1907	0.0056
Si	0.0252	0.0137	0.0219	0.0006
S	0.0277	0.0013	0.0021	0.0001
Ca	0.0243	0.0837	0.1339	0.0033
Fe	0.0214	0.0009	0.0014	0.0000
Sr	0.016	0.0053	0.0085	0.0001
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.1165

Table 5: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 8.97% Borated Polyethylene ($\rho = 1.16 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0668	0.0775	0.0463
B	0.0575	0.0897	0.1041	0.0060
C	0.0502	0.272	0.3155	0.0158
N	0.0448	0.0528	0.0612	0.0027
O	0.0405	0.519	0.6020	0.0244
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.0953

Table 6: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 30% Borated Polyethylene ($\rho = 1.19 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0876	0.1042	0.0623
B	0.0575	0.3	0.357	0.0205
C	0.0502	0.606	0.7211	0.0362
O	0.0405	0.0002	0.0002	0.00001
Si	0.0252	0.0004	0.0005	0.00001
Fe	0.0214	0.0004	0.0005	0.00001
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.1191

Table 7: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 7.5% Lithium Polyethylene ($\rho = 1.06 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0784	0.0831	0.0497
C	0.0502	0.5776	0.6123	0.0307
O	0.0405	0.2613	0.2769	0.0112
Li	0.084	0.075	0.0795	0.0066
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.0983

Table 8: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 78.5% Bismuth-Loaded Polyethylene ($\rho = 2.92 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0309	0.0902	0.0539
C	0.0502	0.184	0.5372	0.0269
Bi	0.0103	0.785	2.2922	0.0236
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.1045

Table 9: Calculations of the Fast Neutrons Effective Removal Cross-Sections for 90% Bismuth-Loaded Polyethylene ($\rho = 3.8 \text{ g cm}^{-3}$)

Element	$\Sigma_R / \rho \text{ (cm}^2 \text{ g}^{-1}\text{)}$	Fraction by Weight	Partial Density $\rho \text{ (g cm}^{-3}\text{)}$	$\Sigma_R \text{ (cm}^{-1}\text{)}$
H	0.598	0.0144	0.0547	0.0327
C	0.0502	0.0866	0.3291	0.0165
Bi	0.0103	0.9000	3.4200	0.0352
Total $\Sigma_R \text{ (cm}^{-1}\text{)}$				0.0845

Table 10: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Borated Silicone ($\rho = 1.59 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	$\Sigma_R(\text{cm}^{-1})$
H	0.598	0.0474	0.0754	0.0451
B	0.0575	0.0108	0.0172	0.0010
C	0.0502	0.1101	0.1751	0.0088
O	0.0405	0.4656	0.7403	0.0300
Na	0.0341	0.0012	0.0019	0.0001
Al	0.0293	0.1875	0.2981	0.0087
Si	0.0252	0.1754	0.2789	0.0070
Fe	0.0214	0.0002	0.0003	0.0000
Zn	0.0183	0.001	0.0016	0.0000
Total $\Sigma_R(\text{cm}^{-1})$				0.1007

Table 11: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Flexi-Boron Shielding ($\rho = 1.64 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	$\Sigma_R(\text{cm}^{-1})$
H	0.598	0.0276	0.0453	0.0271
B	0.0575	0.253	0.4149	0.0239
C	0.0502	0.201	0.3296	0.0165
O	0.0405	0.242	0.3969	0.0161
Si	0.0252	0.269	0.4412	0.0111
Fe	0.0214	0.0041	0.0067	0.0001
Zn	0.0183	0.0026	0.0043	0.0001
Total $\Sigma_R(\text{cm}^{-1})$				0.0949

Table 12: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Borated Hydrogen-Loaded Castable Dry Mix ($\rho = 1.15 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	$\Sigma_R(\text{cm}^{-1})$
H	0.598	0.103	0.1185	0.0708
B	0.0575	0.009	0.0104	0.0006
C	0.0502	0.46	0.5290	0.0266
O	0.0405	0.325	0.3738	0.0151
Mg	0.0333	0.0004	0.0005	0.0000
Al	0.0293	0.0003	0.0003	0.0000
Si	0.0252	0.0043	0.0049	0.0001
S	0.0277	0.0399	0.0459	0.0013
Ca	0.0243	0.0572	0.0658	0.0016
Fe	0.0214	0.0005	0.0006	0.0000
Total $\Sigma_R(\text{cm}^{-1})$				0.1162

Table 13: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Borated Hydrogenated Mix ($\rho = 1.68 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	$\Sigma_R(\text{cm}^{-1})$
H	0.598	0.0337	0.0566	0.0339
B	0.0575	0.0156	0.0262	0.0015
O	0.0405	0.587	0.9862	0.0399
Na	0.0341	0.0059	0.0099	0.0003
Mg	0.0333	0.005	0.0084	0.0003
Al	0.0293	0.239	0.4015	0.0118

Table 13: Contd.,				
Si	0.0252	0.0213	0.0358	0.0009
S	0.0277	0.0019	0.0032	0.0001
Ca	0.0243	0.0883	0.1483	0.0036
Fe	0.0214	0.0027	0.0045	0.0001
Total Σ_R (cm⁻¹)				0.0924

Table 14: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Borated-Lead Polyethylene ($\rho = 3.8 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	Σ_R (cm ⁻¹)
H	0.598	0.0179	0.0680	0.0406
B	0.0575	0.061	0.2318	0.0133
C	0.0502	0.1071	0.4069	0.0204
O	0.0405	0.042	0.1596	0.0064
Si	0.0252	0.0047	0.0178	0.0004
Ca	0.0243	0.0122	0.0463	0.0011
Pb	0.0104	0.8	3.04	0.0316
Total Σ_R (cm⁻¹)				0.1141

Table 15: Calculations of the Fast Neutrons Effective Removal Cross-Sections for K-Resin ($\rho = 1.45 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	Σ_R (cm ⁻¹)
H	0.598	0.1486	0.2155	0.1289
C	0.0502	0.33	0.4785	0.0240
N	0.0448	0.02	0.0290	0.0013
O	0.0405	0.54	0.7830	0.0317
Al	0.0293	0.026	0.0377	0.0011
Total Σ_R (cm⁻¹)				0.1870

Table 16: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Resin 250WD ($\rho = 1.4 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	Σ_R (cm ⁻¹)
H	0.598	0.07243	0.10140	0.06064
B	0.0575	0.01262	0.01767	0.00102
C	0.0502	0.45989	0.64384	0.03232
N	0.0448	0.02086	0.02920	0.00131
O	0.0405	0.33115	0.46360	0.01878
Na	0.0341	0.00031	0.00044	0.00001
Mg	0.0333	0.00055	0.00077	0.00003
Al	0.0293	0.08291	0.11607	0.00340
Si	0.0252	0.00136	0.00190	0.00005
Cl	0.0295	0.00104	0.00146	0.00004
Ca	0.0243	0.01669	0.02336	0.00057
Fe	0.0214	0.00021	0.00029	0.00001
Total Σ_R (cm⁻¹)				0.11816

Table 17: Calculations of the Fast Neutrons Effective Removal Cross-Sections for SUS304 ($\rho = 7.85 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	Σ_R (cm ⁻¹)
C	0.0502	0.0066	0.0518	0.0026
Cr	0.0208	0.18	1.4130	0.0294
Ni	0.019	0.08	0.6280	0.0119
Fe	0.0214	0.7394	5.8043	0.1242
Total Σ_R (cm⁻¹)				0.1681

Table 18: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Krafton-HB ($\rho = 1.08 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	$\Sigma_R(\text{cm}^{-1})$
H	0.5980	0.1066	0.1151	0.0688
B	0.0575	0.0078	0.0084	0.0005
C	0.0502	0.7529	0.8131	0.0408
N	0.0448	0.0220	0.0238	0.0011
O	0.0405	0.1069	0.1155	0.0047
Total $\Sigma_R(\text{cm}^{-1})$				0.1160

Table 19: Calculations of the Fast Neutrons Effective Removal Cross-Sections for Premadex ($\rho = 1 \text{ g cm}^{-3}$)

Element	$\Sigma_R/\rho(\text{cm}^2 \text{ g}^{-1})$	Fraction by Weight	Partial Density $\rho(\text{g cm}^{-3})$	$\Sigma_R(\text{cm}^{-1})$
H	0.5980	0.1140	0.1140	0.0682
C	0.0502	0.4740	0.4740	0.0238
O	0.0405	0.3990	0.3990	0.0162
Si	0.0295	0.0130	0.0130	0.0004
Total $\Sigma_R(\text{cm}^{-1})$				0.1085

CONCLUSIONS

It can be concluded from this work that the selection of a shielding material for fast neutron requires the knowledge of the macroscopic effective removal cross-section (Σ_R). The results obtained from this study can be used as a database for designers of nuclear technology. We can also conclude that the macroscopic effective removal cross-section for fast neutrons (Σ_R) is dependent on chemical content and density of shielding materials.

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